VSEPR as the Thomson Problem: Geometric Necessity of Molecular Geometry from Yukawa Repulsion

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Abstract

For sixty-eight years the valence shell electron pair repulsion (VSEPR) theory has correctly predicted molecular shapes, yet its geometric rules have remained an empirical summary—chemists have known the geometries but not why these geometries are inevitable. Here we show that when valence electron pairs are treated as mutually repelling centres on a sphere, with their effective interaction described by a Yukawa tail potential—the natural form dictated by three-dimensional Euclidean geometry—the energy minimization becomes exactly the generalized Thomson problem. The global minima of the classical Thomson problem for $N = 2$ to 7 are precisely the seven standard VSEPR geometries: linear, trigonal planar, tetrahedral, trigonal bipyramidal, octahedral, and pentagonal bipyramidal. The $N = 5$ trigonal bipyramid is not a Platonic solid; it is acutely sensitive to the functional form of the repulsive potential. A Gaussian overlap potential selects the square pyramid (contradicting the experimental structure of PF_5), whereas the Yukawa potential robustly and uniquely selects the trigonal bipyramid across a wide parameter range. $N = 5$ thus serves as a sharp probe that discriminates the correct repulsive form. $N = 7$ provides a second, independent probe. $N = 4$ and $N = 6$ are Platonic solids protected by group theory and cannot distinguish among different monotonic potentials. This work delivers the first derivation of VSEPR geometry from repulsive-potential minimization, identifies the Thomson problem as the geometric foundation of VSEPR, and shows that $N = 5$ is the sole witness that reveals which repulsive potential nature actually employs.

1 Introduction

In 1957 Gillespie and Nyholm proposed an elegant rule: electron pairs in the valence shell of a central atom repel one another, and the resulting molecular geometry is determined solely by the number of electron pairs, N [?]. Table ?? lists the standard VSEPR geometries. The rule has been taught to every chemistry student since. Yet Gillespie himself acknowledged in 1992 that "there is as yet no rigorous theoretical basis for the VSEPR model" [?]—a candid admission that chemists have relied on a rule whose geometric origin was unexplained.

Table 1: Standard VSEPR geometries

N	Geometry	Bond angles	Example
2	Linear	180°	$BeCl_2$
3	Trigonal planar	120°	BF_3
4	Tetrahedral	109.5°	CH_4
5	Trigonal bipyramidal	90°, 120°	PF_5
6	Octahedral	90°	SF_6
7	Pentagonal bipyramidal	72°, 90°	IF_7

Quantum chemistry has illuminated the mechanism by which electron pairs avoid each other: the Pauli exclusion principle generates an effective repulsion through antisymmetrized wavefunctions [?]; Bader's Atoms in Molecules theory recovers VSEPR geometries from the topology of the electron density and its Laplacian [?]; the Electron Localization Function (ELF) independently confirms that valence electron density concentrates in N distinct angular regions [?]. These approaches explain *why* electron pairs repel—they identify the physical mechanism of avoidance. What they do not explain is the geometric consequence: given that N localized angular concentrations avoid each other on a sphere, *why* do the specific configurations in Table ??—the tetrahedron at 109.5°, the trigonal bipyramid at 90° and 120°, the octahedron at 90°—emerge as the unique energy minima, rather than any other symmetric arrangement?

This paper answers that complementary question. We do not compete with quantum chemistry on the mechanism of repulsion. We take the repulsion as given—electron pairs avoid each other—and ask what geometry necessarily follows. The answer is geometric: when the effective repulsive interaction between electron pairs is modeled with the Yukawa tail potential projected onto a sphere, the global energy minima are exactly the configurations of Thomson's problem [?].

Bent noted the VSEPR–Thomson resemblance as an analogy in 1967 [?], but could only point to the similarity without supplying the physical mechanism that makes the correspondence exact. The present work supplies that mechanism: the total angular energy of N repelling centres on S^2, with a Yukawa pair potential whose form is dictated by the geometry of three-dimensional space. The central result is that the Yukawa tail potential reduces the N-body angular energy minimization to the generalized Thomson problem, and for $N = 2$–7 the known Thomson solutions coincide exactly with the VSEPR geometries.

2 Results

2.1 The geometric origin of the Yukawa form

In three-dimensional Euclidean space, the physical distance between two points on a sphere of radius R separated by an angular interval θ is the chord length $r = 2R\sin(\theta/2)$—not θ itself. Any short-range repulsive interaction that decays exponentially with physical distance must therefore take the form $V \propto e^{-\kappa r}/r$, where κ is an effective screening parameter. This is the Yukawa potential, introduced by Yukawa in 1935 to describe nuclear forces [?] and widely recognized as the mathematical prototype of screened short-range interactions. Projected onto the sphere S^2, the effective angular repulsion between two electron-pair centres separated by θ becomes

$$V(\theta) \propto \frac{e^{-k\sin(\theta/2)}}{\sin(\theta/2)} \tag{1}$$

with the dimensionless parameter $k = 2\kappa R$. This form is not assumed arbitrarily—it follows from the geometry of three-dimensional space. A Gaussian overlap potential $V \propto e^{-\theta^2/4\sigma^2}$, by contrast, implicitly assumes that the exponential decay occurs directly in the angular coordinate θ, ignoring the chord-length projection from $\mathbb{R}^3$ to S^2. This difference is immaterial for highly symmetric configurations but becomes decisive for lower-symmetry cases, as shown below.

2.2 Thomson equivalence

We model each valence electron pair as a repulsive centre on the sphere S^2, with the effective pair interaction given by equation (**??**). The total angular energy is then the sum over all distinct pairs:

$$E(\{\hat{d}_i\}) = E_0 + \sum_{i<j} V(\theta_{ij}) \tag{2}$$

where E_0 collects all configuration-independent self-energy terms and $\theta_{ij} = \arccos(\hat{d}_i \cdot \hat{d}_j)$ is the great-circle angle between the i-th and j-th centres.

For any potential $V(\theta)$ that is strictly monotonically decreasing on $[0, \pi]$, minimizing the total energy E is equivalent to maximizing all pairwise angular separations θ_{ij}—since increasing any θ_{ij} strictly decreases the corresponding $V(\theta_{ij})$. This is the generalized Thomson problem: distributing N mutually repelling points on S^2 to minimize the sum of pairwise repulsions. The classical Thomson problem—with the Coulomb potential $1/r$—has been studied since 1904. Its global minima for $N = 2$–7 are rigorously known [?, ?].

For the Yukawa form (**??**), our numerical optimization across the wide parameter range $k = 0.1$–5.0 yields *exactly the same configurations* as the classical Coulomb Thomson solutions for $N = 2$–7. We therefore have direct numerical evidence that the Yukawa–Thomson problem inherits the same global minima for these N. Table **??** lists these configurations alongside the VSEPR geometries. The correspondence is exact.

Table 2: Thomson global minima = VSEPR geometries

N	Thomson solution	Angular feature	VSEPR	Ref.
2	Antipodal points	$180°$	Linear	[?]
3	Equilateral triangle on great circle	$120°$	Trigonal planar	[?]
4	Regular tetrahedron	$\arccos(-1/3) \approx 109.47°$	Tetrahedral	[?]
5	Trigonal bipyramid	$90°$, $120°$	Trigonal bipyramidal	[?]
6	Regular octahedron	$90°$	Octahedral	[?]
7	Pentagonal bipyramid	$72°$, $90°$	Pentagonal bipyramidal	[?]

The derivation does not require a specific functional form of the electron-pair density profile; it relies only on the monotonicity of the effective potential, which is guaranteed for any localized, unimodal, positive-definite angular distribution of a pair. This form-independence distinguishes the geometric argument from quantum chemical calculations that require explicit choices of basis sets and exchange–correlation functionals.

2.3 $N = 5$ as a probe: Yukawa verified, Gaussian falsified

$N = 4$ (regular tetrahedron) and $N = 6$ (regular octahedron) are Platonic solids whose vertices are inscribed on S^2 with the highest possible symmetry groups, T_d and O_h respectively. Cohn and Kumar proved in 2007 [?] that any completely monotonic potential on S^2 attains its global minimum at these configurations. They are "cheap geometries"—correct for any monotonically decreasing potential, and therefore incapable of distinguishing which potential nature actually employs.

$N = 5$ is not a Platonic solid. The trigonal bipyramid has symmetry D_{3h}, far lower than the octahedral or icosahedral groups. Its energy landscape possesses a competing local minimum (the square pyramid, symmetry C_{4v}) whose relative energy depends sensitively on the precise functional form of $V(\theta)$. This makes $N = 5$ an exceptionally sharp probe.

Table **??** reports the numerical optimization results for $N = 5$ under two different repulsive potentials. The Gaussian overlap potential $V \propto e^{-\theta^2/(4\sigma^2)}$ yields the square pyramid as its unique global minimum. The Yukawa tail potential (**??**) yields the trigonal bipyramid. Experimentally, PF_5 is a trigonal bipyramid [?]. The Gaussian form is falsified; the Yukawa form is verified.

Under the Yukawa potential, the square pyramid is not even a local minimum—starting from the square pyramid configuration with small random perturbations, 10 out of 10 independent relaxations spontaneously converge to the trigonal bipyramid. The Yukawa

Table 3: $N = 5$ discriminates between potentials

Potential	$N = 5$ global minimum	Consistent with PF_5?
Gaussian overlap	Square pyramid	×
Yukawa tail	Trigonal bipyramid	✓

energy landscape for $N = 5$ possesses exactly one minimum: the trigonal bipyramid. This holds across the entire parameter range $k = 0.1$–5.0, covering nearly two orders of magnitude. The trigonal bipyramid is not a fine-tuned outcome—it is the geometrically inevitable result.

$N = 7$ provides a second independent probe. The pentagonal bipyramid (D_{5h}) is also not a Platonic solid. Under the Yukawa potential, optimization uniquely selects the regular pentagonal bipyramid with equatorial angles of 72° and axial–equatorial angles of 90°, matching the structure of IF_7. Under the Gaussian potential, $N = 7$ does not converge to a regular pentagonal bipyramid; the optimized configuration deviates from the experimentally observed regular structure (see Supplementary Information for the full angular output). Two non-Platonic N values, two independent verifications of the same Yukawa form.

2.4 The energy-landscape picture: rigidity versus fluxionality

The angular energy landscape yields geometric predictions that go beyond standard VSEPR. For $N = 4$ and $N = 6$, the energy landscape has a single, isolated global minimum with no competing local minima. These molecules are stereochemically rigid: CH_4 and SF_6 are classical examples.

For $N = 5$, the landscape under the Yukawa potential possesses a unique global minimum (the trigonal bipyramid) with a square-pyramidal saddle point connecting equivalent trigonal bipyramidal configurations. This is exactly the geometric setting of the Berry pseudorotation observed in PF_5 [?]: a low-barrier interconversion that proceeds via a square-pyramidal transition state. The square pyramid is not a stable intermediate but a transition structure—a feature naturally reproduced by the Yukawa landscape. By contrast, the Gaussian potential erroneously stabilizes the square pyramid as a local minimum, which would predict a different fluxional behaviour inconsistent with experiment.

For $N = 7$, the Yukawa landscape supports near-degenerate minima that enable the fluxionality observed in IF_7. The present model thus provides an energy-landscape-based criterion for rigidity versus fluxionality directly from the repulsive potential.

2.5 Transferable lone-pair to bond-pair width ratios

When lone-pair electrons are present, the VSEPR framework assigns them a larger angular width than bond pairs, reflecting the absence of a confining opposing nucleus. We encode this as $\sigma_{\mathrm{lone}} > \sigma_{\mathrm{bond}}$ in a field-patch description. Calibrating the lone-pair to bond-pair width ratio from the NH_3 bond angle (106.7°) and applying it without adjustment predicts the NCl_3 bond angle to within 0.4° and NH_2CH_3 to within 0.9°. For phosphorus, calibration from PH_3 predicts PF_3 to within 0.7° and $P(CH_3)_3$ to within 0.1°. The transferability holds within a given element and period, consistent with an atomic origin of the patch width. Cross-period trends show an increase in effective σ_{lone} for heavier elements, attributable to core-shell screening.

3 Discussion

The identification of VSEPR geometries with Thomson configurations is exact, physically motivated, and robust. It does not compete with quantum chemical explanations of electron-pair repulsion—it complements them. Quantum mechanics answers *why* electron pairs repel; the Thomson equivalence answers *what geometry* necessarily results from that repulsion on a sphere. Together they provide a complete picture: Pauli exclusion and electron correlation generate the effective repulsion, and the geometry of three-dimensional space channels that repulsion into the Thomson minima.

The Yukawa form is not an arbitrary choice—it is dictated by the geometry of three-dimensional Euclidean space. Any short-range interaction between localized entities decays with their physical separation r, not their angular separation θ. Projecting this decay onto the sphere necessarily produces the $\sin(\theta/2)$ structure of equation (**??**). The Gaussian form, by imposing exponential decay directly in θ, violates this geometric constraint. In the highly symmetric $N = 4$ and $N = 6$ configurations, the violation is masked by group-theoretic protection: any monotonic potential yields the same global minima because the Platonic symmetries are simply too rigid to be perturbed. The violation is exposed only at $N = 5$, where the lower symmetry makes the energy landscape sensitive to the precise functional form. This explains why VSEPR could not be rigorously derived for sixty-eight years: the correct potential had to be identified, and $N = 5$ is the only witness that can tell the correct potential apart from convenient mathematical approximations.

Our derivation carries an honest boundary. The strict analytic proof that the effective $V(\theta)$ is monotonically decreasing for arbitrary localized electron-pair profiles remains an open mathematical problem; we have verified it numerically for three qualitatively distinct profile functions—Gaussian, exponential, and compact-support—covering the range of physically plausible shapes. The quantitative accuracy of cross-period lone-pair width predictions is limited by the saturation of the search upper bound from the third period onward; true values are likely larger. Strongly electronegative ligands (e.g., fluorine) narrow bond-pair patches and distort angles beyond the minimal model, and bulky ligands widen them; these effects are identifiable extensions for future work. None of these limitations affect the central result: the Thomson–VSEPR equivalence derived from the Yukawa potential.

4 Methods

All numerical optimization was performed on the unit sphere S^2 using the SLSQP algorithm implemented in SciPy (Python 3.x). For each $N = 2$–7, 50–300 random initial configurations were generated and optimized independently to locate the global minimum. The Yukawa potential was evaluated with $k = 1.0$; robustness was verified over the range $k = 0.1$–5.0. The Gaussian comparison potential was evaluated with $\sigma = 0.3$. For $N = 5$ relaxation tests, a square pyramid configuration was perturbed with random angular noise (amplitude 0.2 rad) and re-optimized 10 times; all trials converged to the trigonal bipyramid. Cross-period calibration of the lone-pair to bond-pair width ratio was performed on NH_3, H_2O, PH_3, H_2S, AsH_3, H_2Se, SbH_3, and H_2Te, with 30–40 trials each.

Data Availability

All numerical data reported in this paper can be independently reproduced using the open-source Python code deposited at Zenodo (DOI to be inserted upon acceptance). The Thomson solution angles are independently verifiable from the published mathematical literature [?, ?].

Code Availability

The optimization and analysis code is available at Zenodo (DOI to be inserted). No proprietary software was used.

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Supplementary Information

The Supplementary Information (SI) contains:

- Full angular output (pairwise angles) for Gaussian $N = 7$ and Yukawa $N = 7$ global minima, side by side.
- Energy convergence plots for $N = 5$ under both potentials.
- Tabulated lone-pair calibration data across periods.

These data allow direct inspection of the raw geometric output and verify the statements in the main text concerning the Gaussian $N = 7$ deviation and the Yukawa $N = 7$ regularity.

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